

Answer **all** questions.

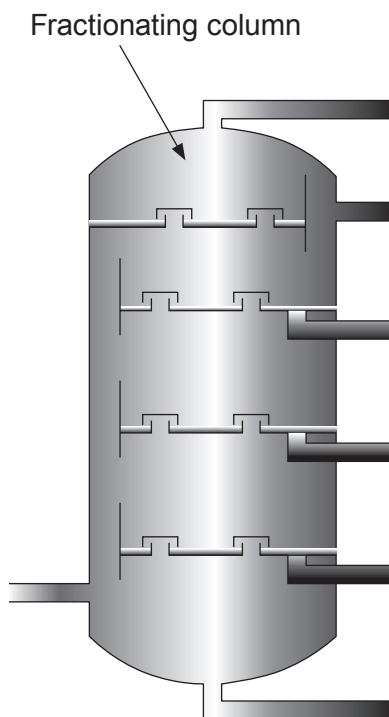
1. Raw materials are extracted from the Earth’s crust or atmosphere by different methods.
- (a) Identify the correct extraction method for each raw material by drawing a line.
One has been done for you.

[2]

Raw material	Extraction method
crude oil	fractional distillation
shale gas	drilling
haematite (a metal ore)	surface mining
oxygen	fracking

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- (b) A fractionating column is used to separate the hydrocarbons in crude oil. The table shows information about some of the fractions obtained.



Fraction	Carbon chain length	Boiling range (°C)
Refinery gases	1-4	<40
Petrol	5-8	40-110
Naphtha	8-10	110-180
Kerosene	10-16	
Diesel	16-20	260-320
Fuel oil	20-50	320-400
Bitumen	>50	>400

- (i) **Complete** the table. [1]
- (ii) Name the process which is used to separate crude oil into its fractions. [1]
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- (iii) State what happens to the crude oil immediately before it enters the column. [1]
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- (iv) State the fraction that a hydrocarbon with a boiling point of 290 °C will belong to. [1]
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(c) Complete the statements below by underlining the correct term in the brackets.

(i) The temperature (**increases, stays the same, decreases**) going up the fractionating column. [1]

(ii) The boiling point of a hydrocarbon (**increases, decreases, stays the same**) as the chain length increases. [1]

(d) The fractionating column produces hydrocarbons that can be further processed and used to make plastics.

Complete the table to show how polyethene, polychloroethene (PVC) and polytetrafluoroethene (PTFE) are made. [3]

Monomer			Polymer	
Name	Molecular formula	Structural formula	Name	Structure
Ethene	C_2H_4	$ \begin{array}{c} H & & H \\ & \diagdown & / \\ & C = C \\ & / & \diagdown \\ H & & H \end{array} $	polyethene	
Chloroethene	C_2H_3Cl		PVC	$ \left[\begin{array}{cc} Cl & H \\ & \\ -C & -C- \\ & \\ H & H \end{array} \right]_n $
Tetrafluoroethene		$ \begin{array}{c} F & & F \\ & \diagdown & / \\ & C = C \\ & / & \diagdown \\ F & & F \end{array} $	PTFE	$ \left[\begin{array}{c} F \\ \\ -C- \\ \\ F \end{array} \right]_n $